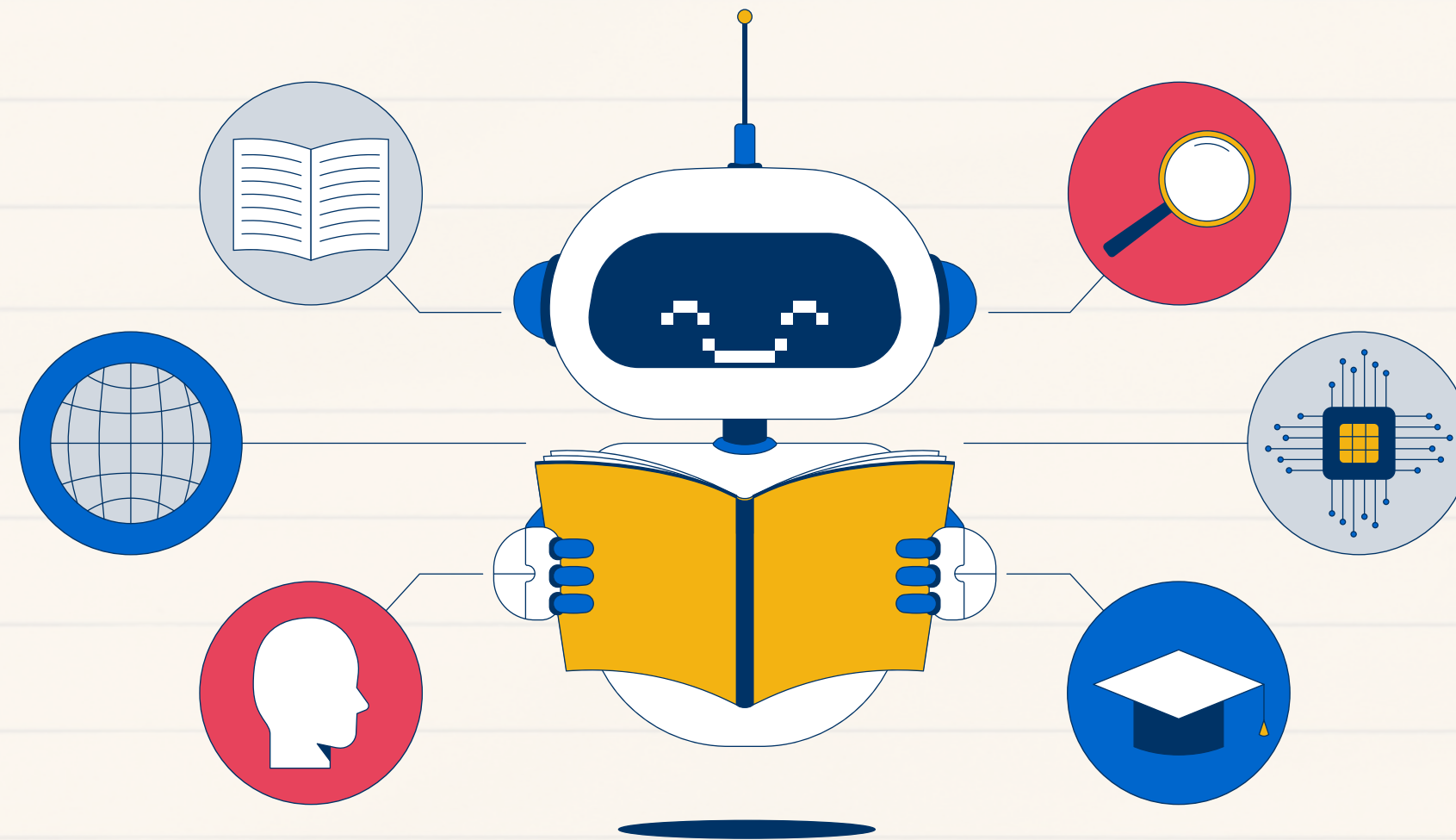


BuildBot Project



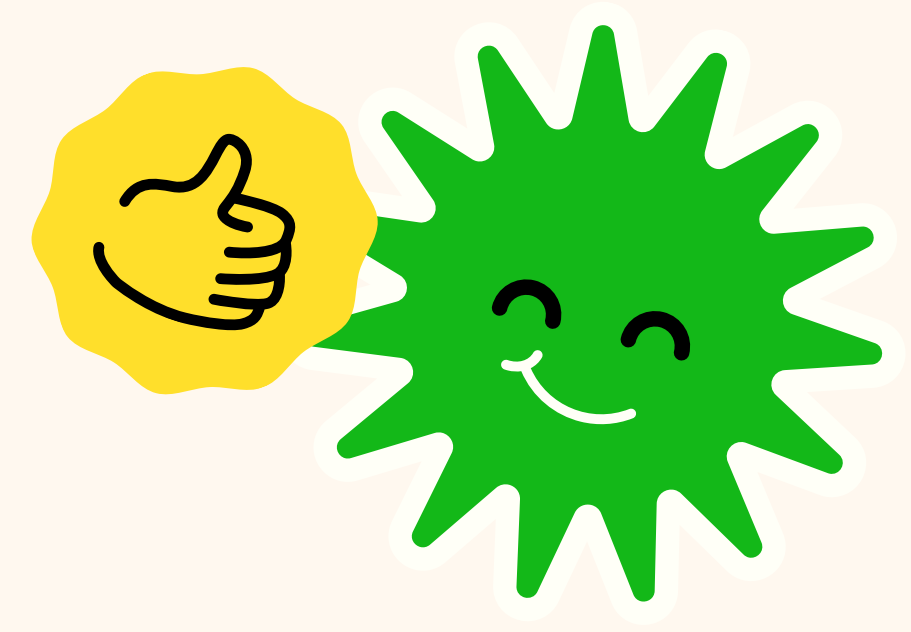
Week 1

Left to do on W1 materials

- select marty presentaion video
- translate video
- QR codes

TODAY'S AGENDA

Week 1: Introduction to social robots



7:25 - 7:30

Project presentation

7:30 - 8:30

Research related test

8:35 - 8:35

Intro to Robots and Marty

8:40 - 9:35

Building Marty - legs

9:35 - 9:45

Feedback



Introduction of the research team



Luca M. Leisten
PhD student
ETH Zurich



Olga Muss
Master student
University of Neuchâtel

Participating in a research project

The research team



- prepare and carry out the research
- conduct pre and post tests
- assist teachers during classes

Luca Leisten & Olga Muss

Teachers

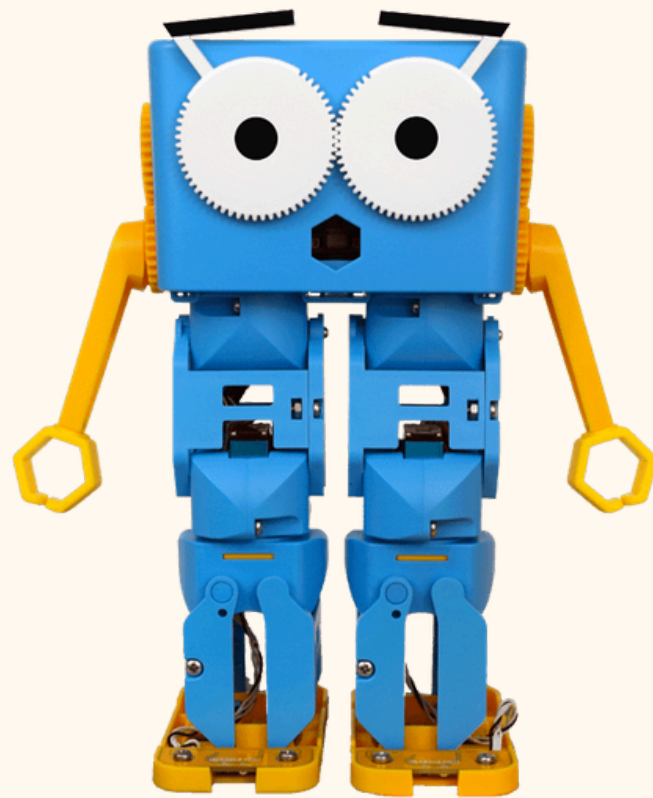
- help with materials preparation
 - teach the classes
 - assist the students
-

Students

- do the pre and post tests
 - participate in class
-

This is a new topic for teachers too, so we will learn together

We are going to build a social robot, design how it will interact and then test it



Marty

Step 1: Build Marty

Step 2: Let Marty move

Step 3: Make Marty autonomous

Step 4: Let Marty perceive and speak

Step 5: Design safe interaction with Marty

Step 6: Test Marty's learning use

6-week program

	Theme	Activity
Week 1	Social robots	Build
Week 2	Artificial Intelligence (AI) and Large Language Models (LLMs)	
Week 3	Limits of AI	
Week 4	Programming and Arduino	Programming and Interacting
Week 5	Sensors and large language models (LLMs)	
Week 6	Large Language Models (LLMs) Design and Interaction	

Short questionnaires after each lesson,
plus long questionnaires in weeks 1, 3, 6 and after a few weeks

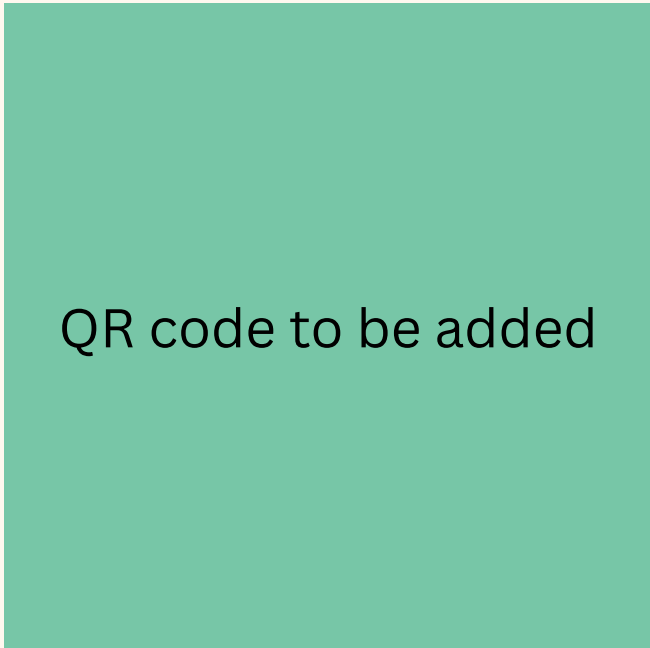
[Access the Research
Project Student guide](#)



Researchers will take photos and videos. You will only be recognizable in them if you agree at the beginning of the questionnaire.

Consent Form

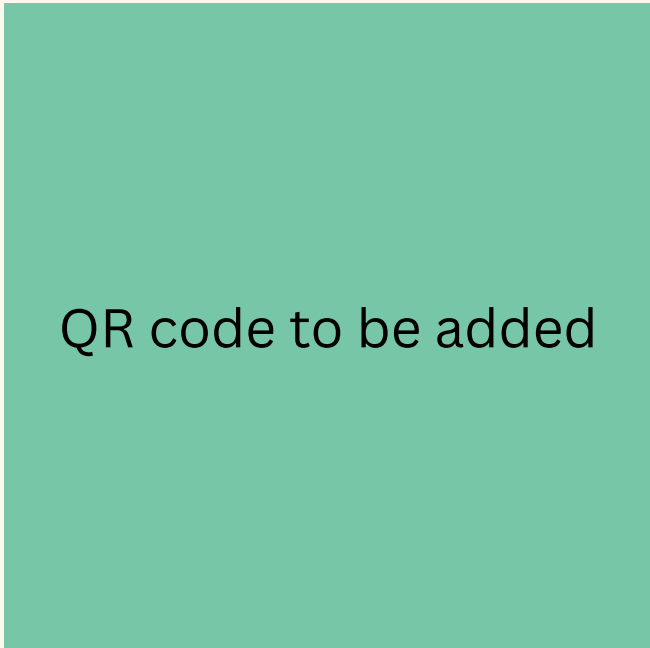
It contains the presentation of the project and what it means to participate in this study. In a form there, you can indicate if you agree to participate in the study and if you agree that we take pictures of you.



QR code to be added

Pre-test

There is no judgement of your individual answers, researchers look at trends.
All data will be anonymous.



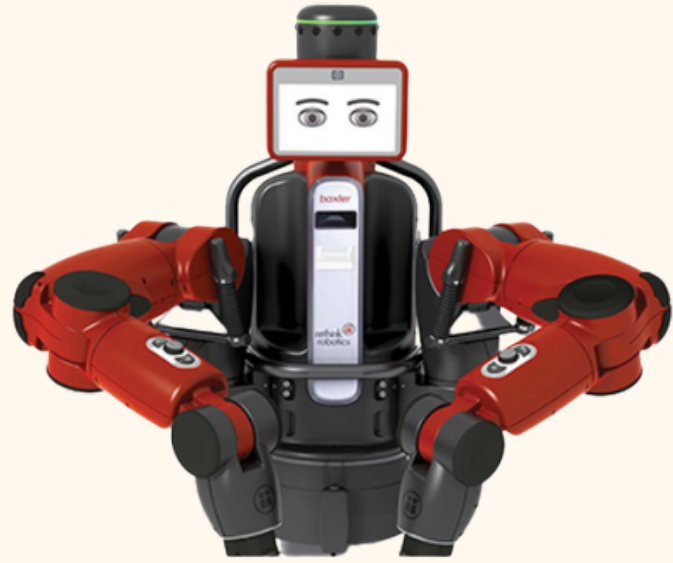
QR code to be added

We will be working in groups of 3 or 4

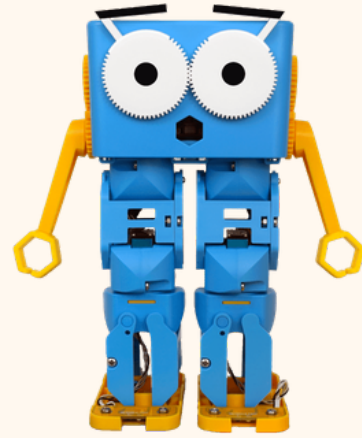
Groups

Do you all know each other? If not, introduce yourself quickly.

Introduction to Social Robots



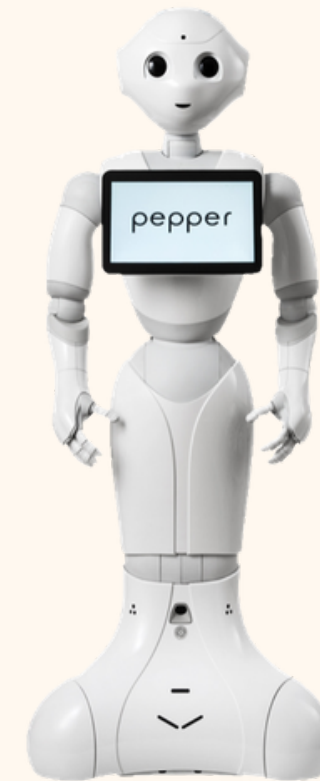
Baxter



Marty



Nao



Pepper



Cosmo



Paro



Blossom

Activity: choose one robot per group, do 5 minutes research what these robots are used for and what they can do, then present briefly to the class ¹¹

There are many types of robots

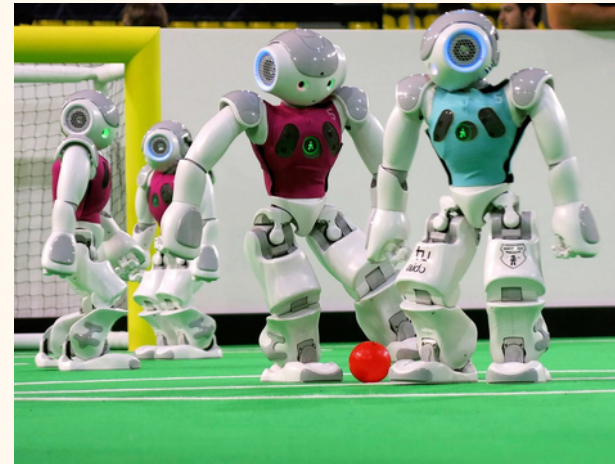
Industrial robots



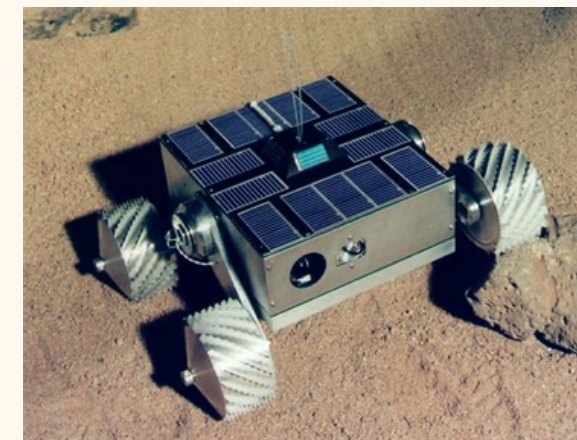
Service robots



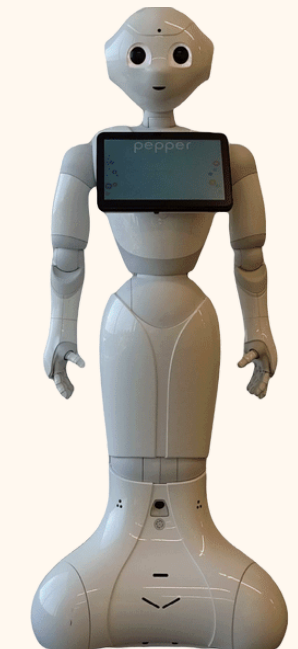
Entertainment robots



Exploration robots



Social robots



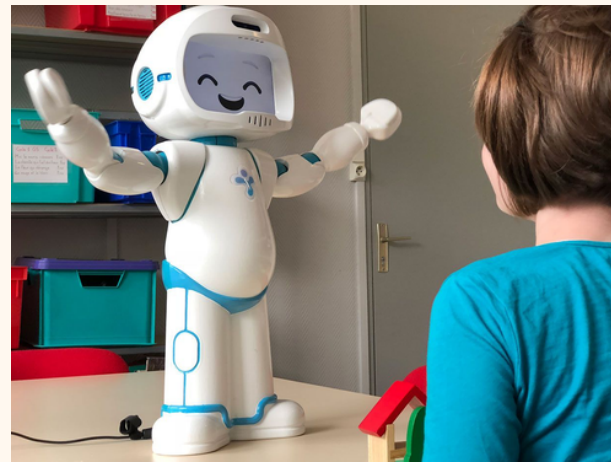
and many other... 12

Social robots are physically embodied agents with some autonomy, able to engage in social interaction, communication, cooperation, and decision-making with humans. These capabilities are interpreted as ‘social’ by people, according to cultural norms and conventions. Henschel et al. (2021)

Healthcare



Education



Customer service



Entertainment

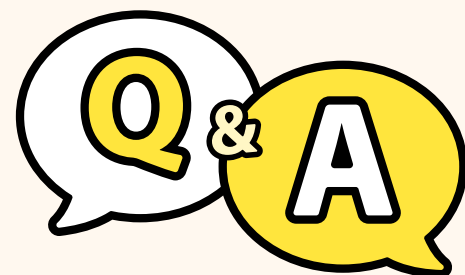


Assistance



these are examples, there are many more

Generally, the goal is not to replace humans, nor to become friends with humans, but to assist.

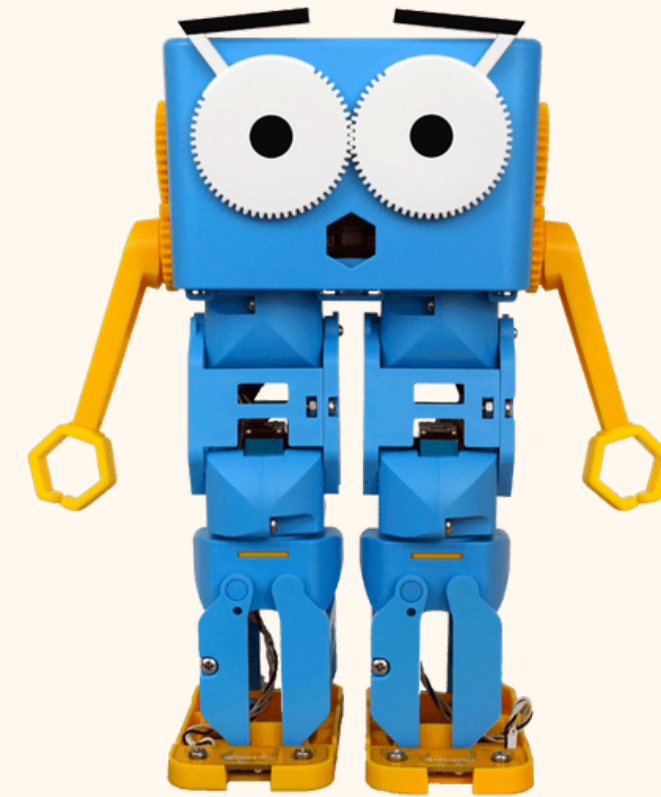
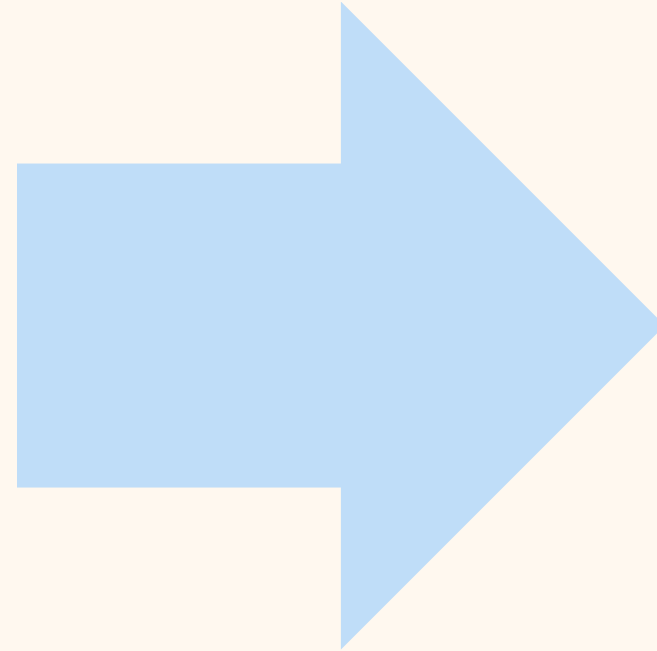
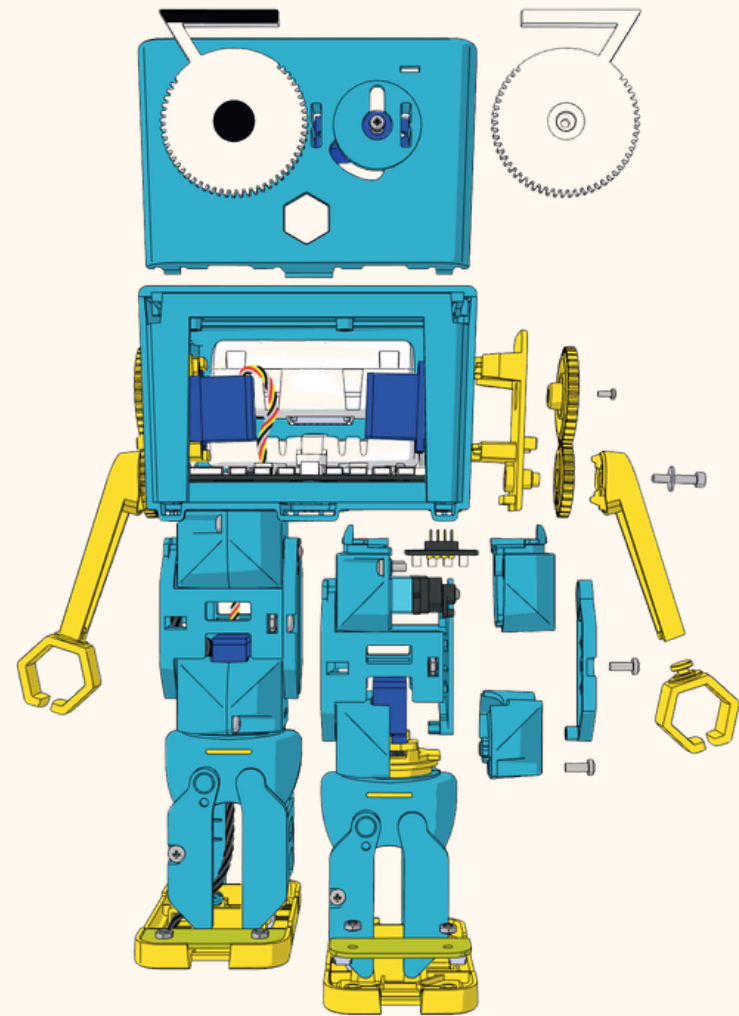


What characteristics should a robot have to be considered a social robot?

Using Marty Robot from Robotical



Building Marty



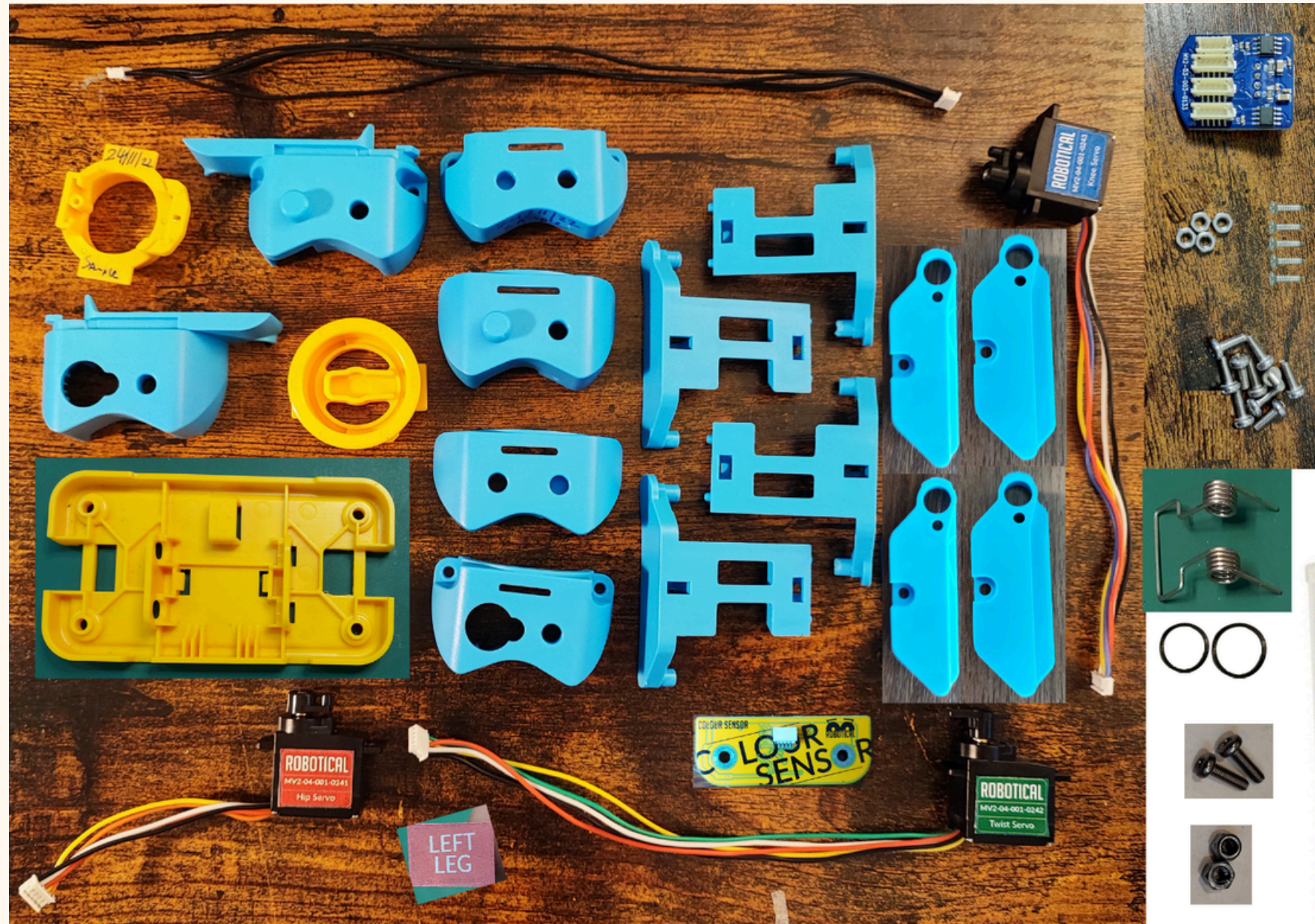
Marty

[Access the Building Guide](#)

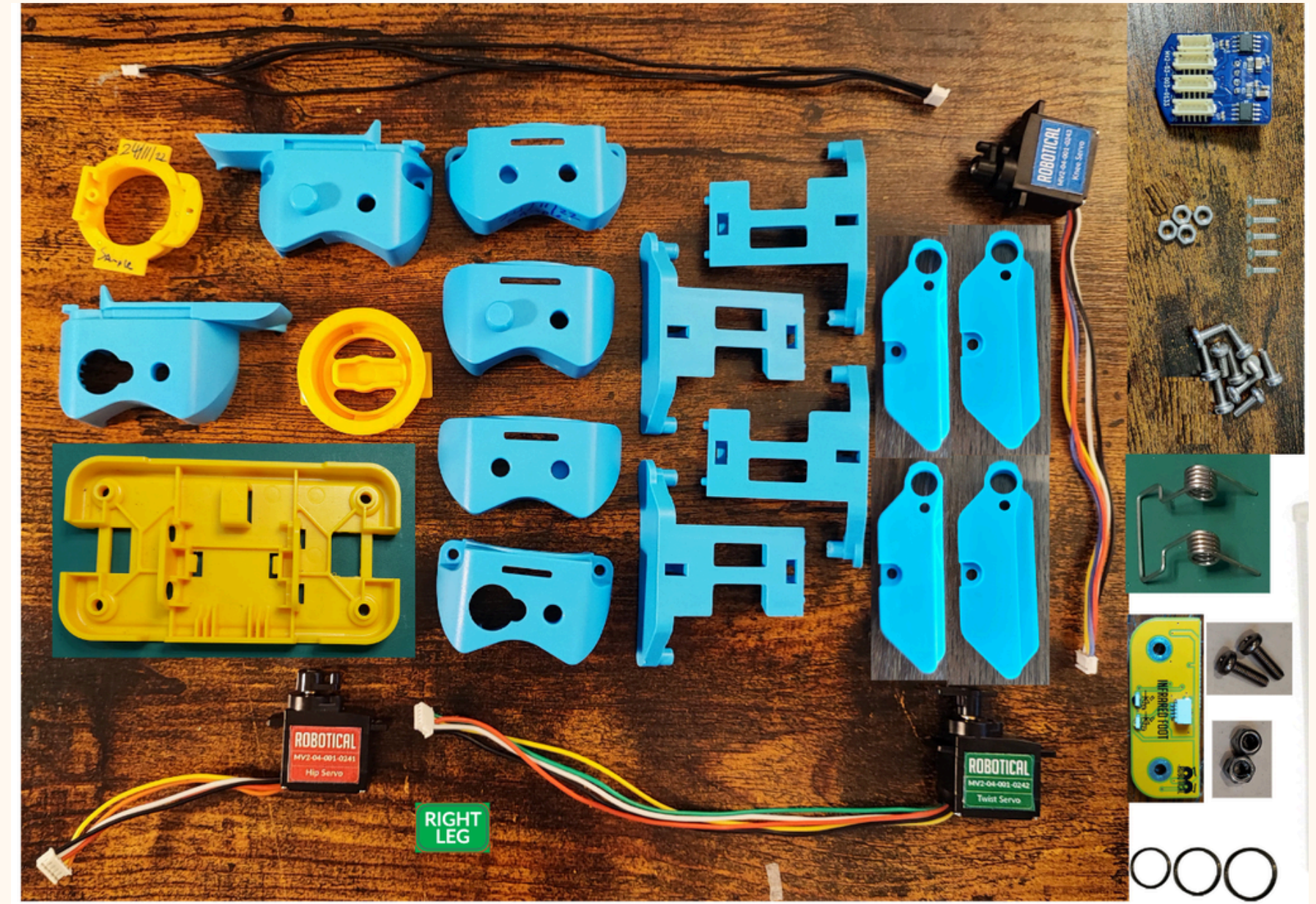


Today, we are building the legs

left leg parts



right leg parts



Organize your work: is everyone working on one leg? are you working in parallel on 2 legs?

Pay attention the 2 legs are different: different motors, sensors, a few pieces.

There are some special parts

Motors



each leg

Hip motor - lifts the whole leg

Twist motor - rotates lower leg at knee

Knee motor - bends the knee

Sensors

Right leg



Infrared Sensor - detect objects in-front and underneath the foot

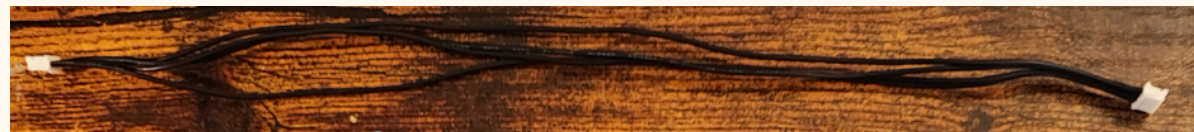
Left leg



Color Sensor - recognizes colors

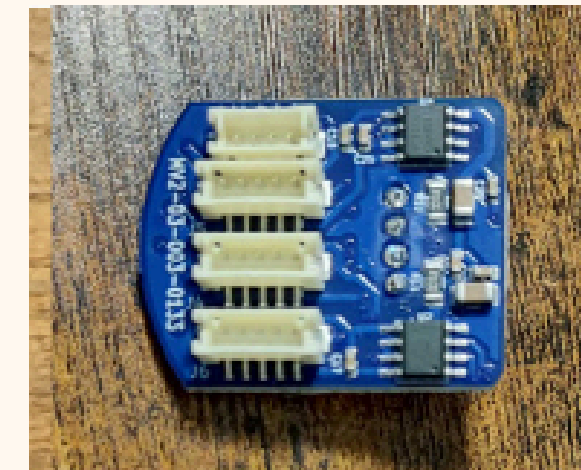
Cables

each leg



Cable connecting foot sensor to the main board in the head

Other



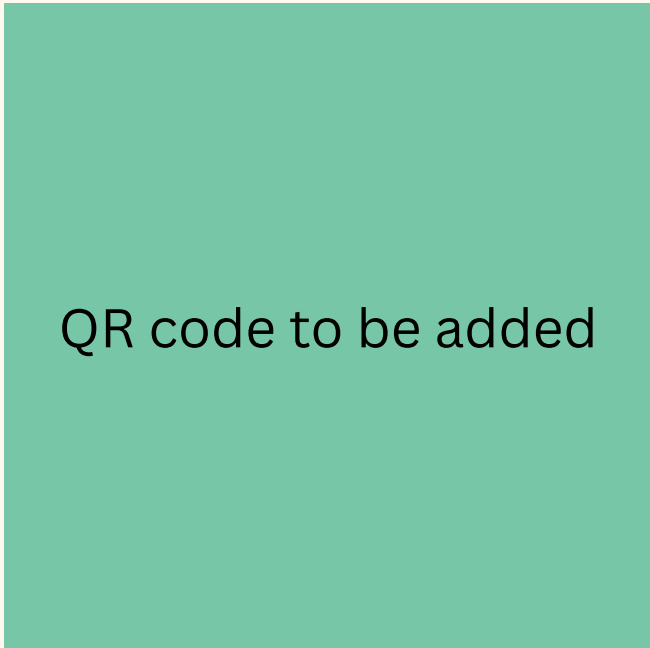
each leg

Printed Circuit Board (PCB) connecting the leg motors and sensor to the main board in the head

You may start

Feedback

Please fill in the survey online



QR code to be added

Annexes

Learning goals

Week 1: Introduction to social robots

15 minutes

Presenting the project:

- Definition of a social robot and how it is different from other robots and virtual agents
- Projects goal and timeline

Presenting Marty

- Who is Marty
- What it can do
- What we will achieve